



characteristics and the unique design challenges and opportunities they present.

Folding phones, such as Samsung's Galaxy Z Fold series and Huawei's Mate Xs, are innovative devices that combine the compactness of a smartphone with the spacious display of a tablet. On the other hand, big screens refer to devices like tablets, which offer larger display areas compared to traditional smartphones. Some devices even blur these lines, such as the Microsoft Surface Duo, which features two screens hinged together.

These devices are hallmarked by their larger display areas and unique screen configurations. Single-screen foldable devices feature one continuous display that can be folded, while dual-screen foldables consist of two separate screens joined by a hinge. In both cases, users can run multiple apps simultaneously on the same screen using multi-window mode. Moreover, these devices can adopt different postures depending on how the user holds or positions the device, such as flat, half-opened, or in a tabletop mode where part of the screen is flat, and part is tilted towards the user.

The advent of these devices presents a new set of design challenges for app developers. The varied screen sizes, postures, and the ability to run multiple apps simultaneously necessitate a rethink of traditional UI and UX design principles. Furthermore, certain technical considerations, such as handling screen continuity when transitioning between folded and unfolded states, require additional attention.

However, these challenges also open up a wealth of opportunities for innovation in app design. The larger display areas and versatile screen configurations enable richer, more immersive experiences and enhance productivity applications. For instance, apps can display more information at once or use multi-column layouts to make better use of the extra screen space.

The rise of folding phones and big-screen devices also brings changes to user behavior. These devices lend themselves well to multitasking, with the ability to run and view multiple apps simultaneously. For example, a user could have a video call run-

ning on one half of the screen while taking notes or browsing the web on the other.

Moreover, these devices elevate media consumption experiences. The larger screens offer more immersive viewing experiences for videos and games, while the ability to fold the device offers flexible viewing angles. Some apps, such as YouTube Music, have been optimized to enhance the user experience on these larger screens, offering features like side-by-side views of player controls, lyrics, and related content.

Case studies: Apps optimised for folding phones and big screens

The advent of foldable phones and larger screens has inspired app developers to optimize their offerings for these novel devices. This section will delve into case studies of three apps that have successfully adapted to the unique characteristics of foldable phones and big screens: Google Calendar, YouTube Music, and Calm.



A. Google Calendar

The Google Calendar app has been redesigned with a fully adaptive layout, enabling users to leverage the larger screen space offered by foldable phones and large-screen devices. The app now presents a split-screen view: the monthly calendar on the left and details of the day's events on the right. This feature allows users to view a broad overview and specific details simultaneously, enhancing productivity and user experience. A new navigation bar at the top further facilitates easy switching between different views, such as day, week, or month. The transition from the front screen of a foldable phone to the inner screen is seamless, demonstrating effective use of app continuity features. By making these changes, Google Calendar has successfully optimized its app for foldable and big-screen devices, improving functionality and user experience.

B. YouTube Music

YouTube Music provides an immersive and optimized listening experience on large screens. The app has been redesigned with a two-column layout, which enhances the playlist experience.